GIS TERM PROJECT · FINAL PRESENTATION

Geospatial Reasoning using LLMs and Generative AI

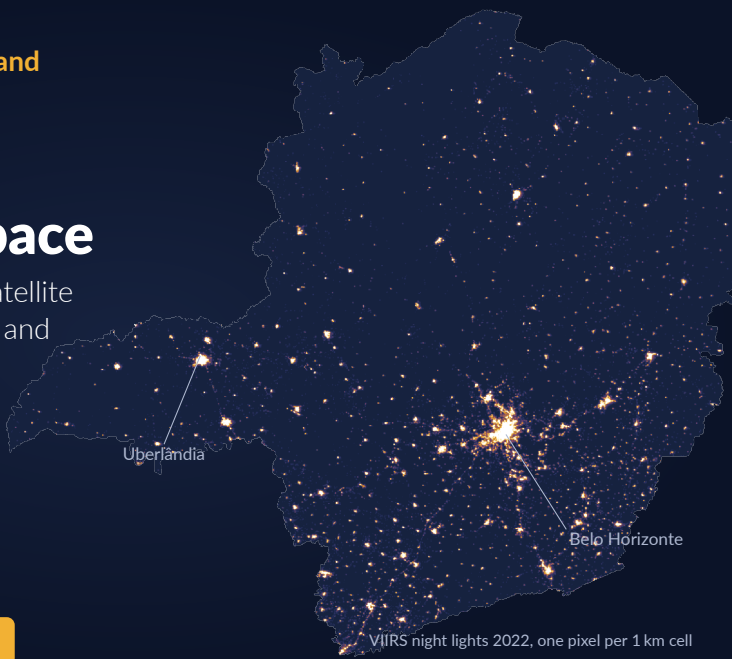
Reading the economy from space

Estimating municipal GDP from satellite embeddings, text and night lights, and putting it to use

Group 2

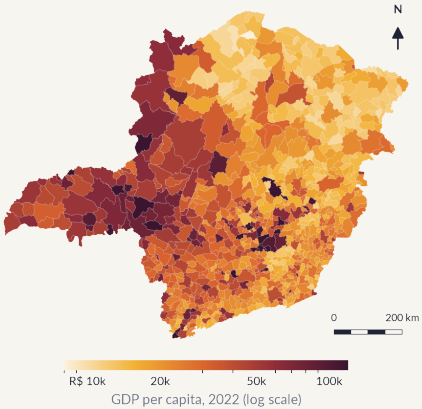
Hemant (22CS30029)
Saras Dipak Wagh (22CS30048)
Pallav Agarwal (22CS30040)

github.com/chadsaras/GIS_Project



VIIRS night lights 2022, one pixel per 1 km cell

Local GDP is patchy; satellites see every place



49×

gap in GDP per capita inside one state:

Catas Altas (R$ 391k) **vs São João das Missões** (R$ 8k)

- Sub-national GDP is published **late**, **estimated** from partial data, or **missing** where statistics are weak.
- Satellites measure **every place, every year, with the same sensor**.

GOAL

Estimate GDP per capita for all **853 municipalities** of Minas Gerais from **free satellite data**, test it **honestly**, and turn it into a **usable tool**.

Since the mid-term: from first results to a tested application

AT THE MID-TERM (6 OCT)		FINAL (NOVEMBER)
DONE Models V1–V12, RQ1–RQ4 answered	→	NEW **Same answers after a full rerun** from scratch
RUNNING Caption audit	→	DONE **35% of AI captions** hold a false claim; checks cut it to 23%
CODE READY Error analysis (RQ5)	→	DONE **Mines, steel mills, commuter towns** are invisible from space
MISSING Uncertainty	→	NEW **Honest 90% ranges** (89.6% held) and a risk warning
MISSING Test outside the study area	→	NEW **São Paulo transfer**: satellites travel, night lights do not
MISSING Application	→	NEW **Web application**: estimates where statistics are missing

Also: a 12-page report, a one-command rerun of the whole pipeline, and checks of ideas that did not work (slide 16).

Two recent ideas, never tested together

UrbanCLIP Yan et al., WWW 2024

A language model writes a **caption for every satellite tile**; image and text are aligned.

– GDP gain only +0.5% R^2; random split lets neighbours leak into the test

Satellite embeddings Yue, Zhao & Hu, 2026

Google's **64-number embeddings**, **steered by night lights** towards economic meaning.

– countries only; no text, no POI or roads

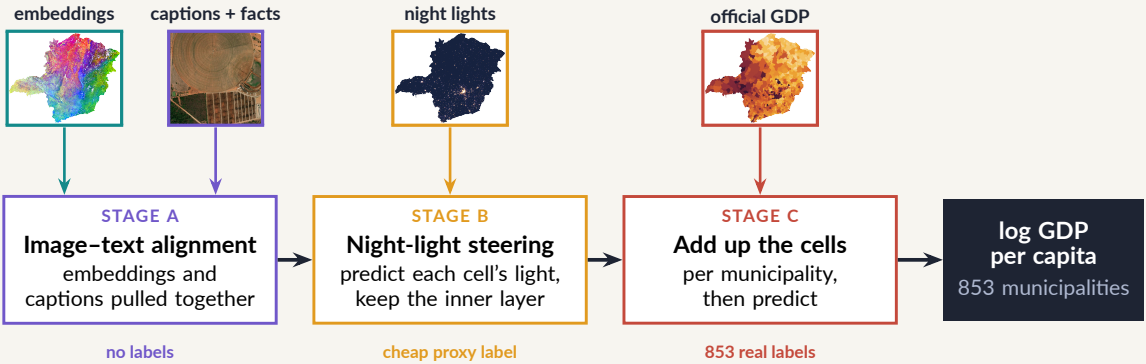
THIS PROJECT

Both, in one pipeline

● embeddings ● text
● night lights ● POI, roads

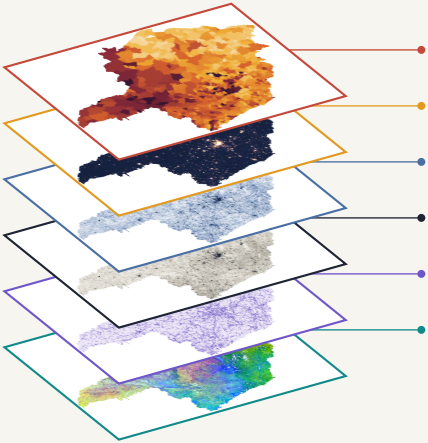
853 municipalities · spatial test ablation of every ingredient

Three learning stages, from no labels to official GDP



Captions are needed only in training: at prediction time each cell uses its embedding alone, so any place can be estimated.

Six free layers on one 1 km grid



GDP per capita (target)
IBGE 2022, official

Night lights
VIIRS 2022

Roads and POIs
OpenStreetMap

Buildings
Microsoft + OSM footprints

Captioned tiles
10,000 Esri tiles, Qwen2.5-VL

Satellite embeddings
Google GSED, 64 numbers, 10 m to 1 km

853
municipalities

575k
1 km cells

10,000
captioned tiles

10M
building footprints

All layers free and aligned on the same grid (EPSG:31983). The target is official statistics, not night lights.

AI captions help, but a third of them contain a false claim



Esri World Imagery, about 1 m per pixel

QWEN2.5-VL-7B CAPTION, AFTER THE CHECKS

- ✗ ~~The satellite image shows a large circular area with concentric rings, likely a mine pit, surrounded by a grid of smaller rectangular plots of farmland.~~ **no industry in the data**
- ✓ Roads crisscross the area, connecting the plots and the perimeter.
- ✓ Vegetation is sparse, mostly along the edges and in some patches within the farmland.
- ✓ A small body of water, possibly a pond, is visible near the farmland.
- ✓ No buildings or industrial structures are evident, suggesting minimal built-up area.
- ✗ ~~The overall scene appears to be an agricultural landscape with a focus on farming and possibly mining operations.~~ **filler**

Checks against land cover, roads and building footprints keep 84% of 60,023 sentences.

BLIND AUDIT OF 300 CAPTIONS

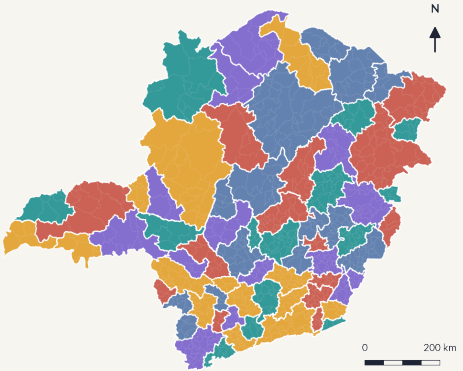


captions with ≥1 false claim, 95% interval

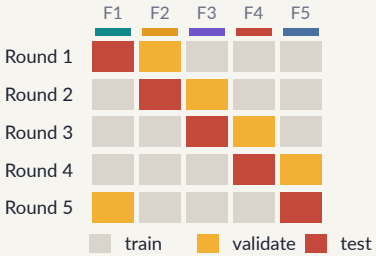
Cleaning removes **41%** of false sentences; **58%** of what it removes is false or vague.

Labelled by an LLM (Claude) from each tile; a human check is still open.

Every municipality is tested in a region never seen



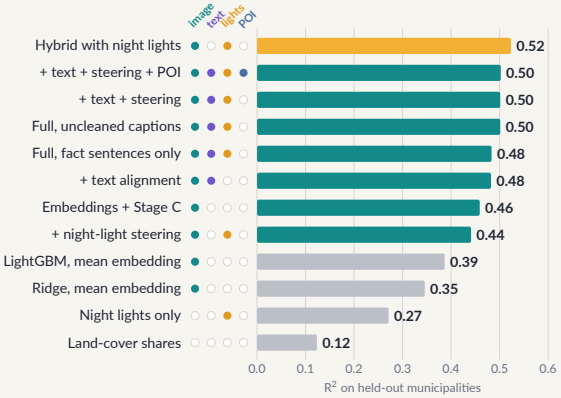
70 official regions (white outlines) grouped into 5 folds



- Whole regions held out: **no look-alike neighbour** in training. All stages retrained each round.
- The same rule holds for every range, warning and choice shown later: **nothing is tuned on test towns**.

Models are compared on the same folds with a block bootstrap over regions (95% intervals).

Satellites explain half of GDP gaps, twice night lights



Dots: ingredients each model uses. Hybrid = best model blended with night lights.

0.52 R^2, hybrid, unseen regions

CONFIRMED BY A FULL RERUN

RQ2 TEXT **small gain, not robust**

ΔRMSE −0.025, rerun −0.009; intervals include 0

RQ1 STEERING **no clear effect**

ΔRMSE +0.007, 95% CI [-0.028, +0.041]

RQ3 POI, ROADS **nothing added**

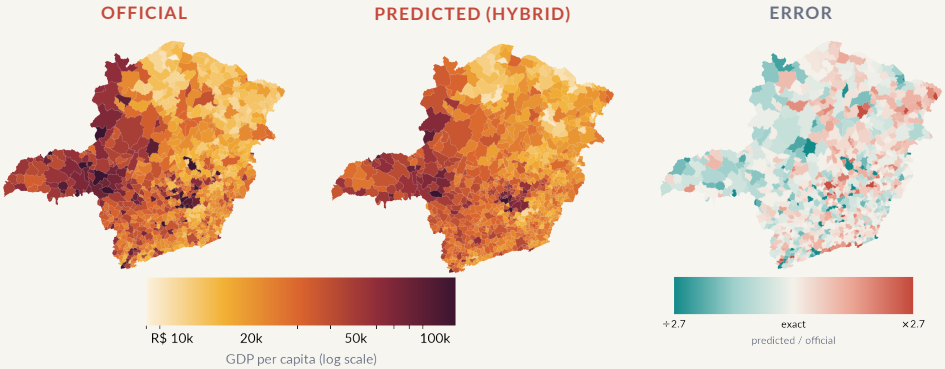
ΔRMSE -0.001, 95% CI [-0.017, +0.016]

RQ4 HYBRID **best, not significant**

ΔRMSE -0.009, 95% CI [-0.035, +0.015]

RMSE differences (log GDP per capita), 95% intervals; negative = first model better.

Predictions follow the map, but are pulled towards the middle



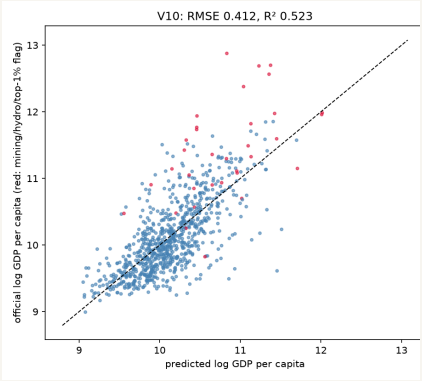
75%

of municipalities predicted within a factor 1.5 of official

Under-predicted: rich agribusiness and mining towns (west, centre).

Over-predicted: small public-sector and commuter towns.

The biggest misses are mines and steel mills satellites cannot see



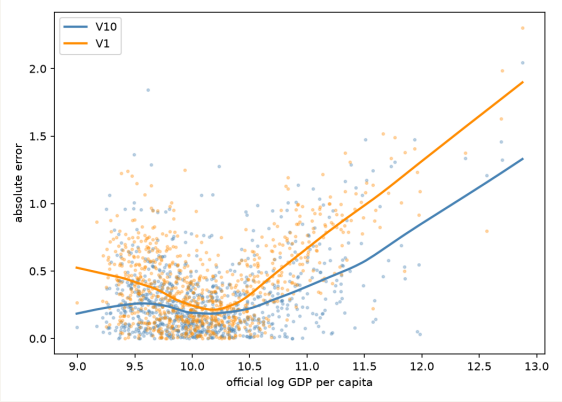
Hybrid, 853 test predictions; red = flagged mining or hydro.

TEN LARGEST GAPS

Municipality	gap	likely reason
Catas Altas	−2.04	iron-ore mine
Santa Cruz de Minas	+1.84	tiny commuter town
Ressaquinha	−1.48	large plant
Conceição do Mato Dentro	−1.48	iron-ore mine
S. Gonçalo do Rio Abaixo	−1.46	iron-ore mine
Mário Campos	+1.37	BH commuter town
Itatiaiuçu	−1.34	iron-ore mine
Extrema	−1.32	logistics hub
Ouro Branco	−1.31	steel mill
Chiador	+1.29	public-sector town

Gap = predicted − official (log). A few cells make most of the output. Reasons are candidates.

Errors grow with income, but half as fast as with night lights



Absolute error against official income, LOWESS. Blue: hybrid; orange: night lights only.

0.08 vs **0.19**

extra error per log point of income, hybrid vs night lights

0.16 vs **0.33**

Moran's I of errors: still clustered, half as much

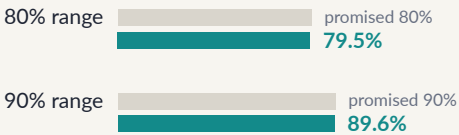
$\rho = 0.04$

spread across training seeds does not predict error: seeds are not a confidence interval

Error regression with region-clustered SEs: errors rise with income and built-up share, fall with services share; OSM completeness does not matter. This is why the application needs its own honest ranges (next).

Every estimate ships with an honest range and a warning

RANGES: PROMISED VS ACHIEVED



Share of held-out towns whose official value fell inside the range. Ranges are set from past errors on validation towns (split conformal), so they need no extra model.

EXAMPLE

Catas Altas (iron mine): estimate R$ 51k, range 21k–122k, HIGH RISK. Official: R$ 391k. The tool cannot see the mine, but it says so.

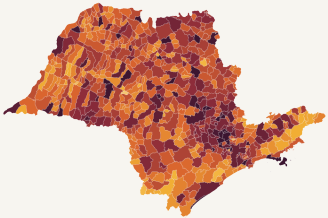
RISK LABEL FROM SATELLITE SIGNALS ALONE

Risk	typical error	mining/hydro towns	90% range
● Low (299 towns)	×/÷ 1.27	2	×/÷ 1.61
● Medium (263)	×/÷ 1.34	5	×/÷ 1.87
● High (291)	×/÷ 1.45	**27 of 34**	×/÷ 2.42

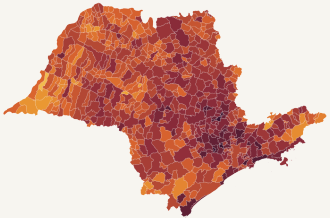
Signals: bare ground (open pits) and a high estimated income. Every group's range still holds about 90%. The warning ranks risk moderately well (AUC 0.66): it tells you where not to trust a number; it does not fix it.

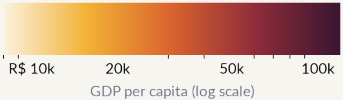
Trained in Minas Gerais, it still works in São Paulo

OFFICIAL, IBGE 2022

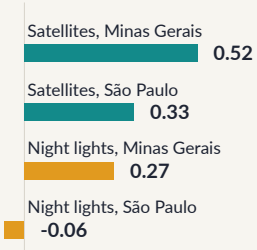


PREDICTED, NEVER TRAINED ON SP



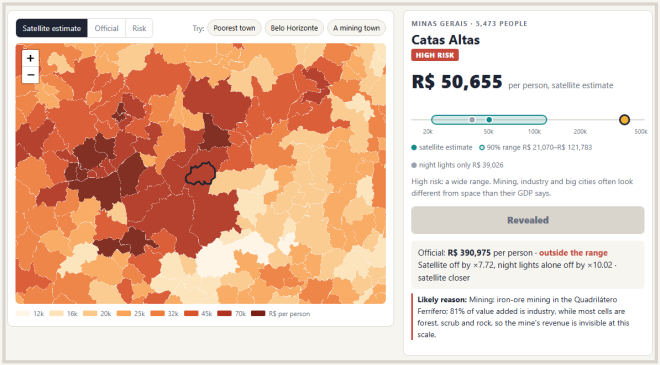


R^2 AGAINST OFFICIAL GDP



645 municipalities, models unchanged. Rank agreement **0.61** (night lights 0.33). Minas Gerais 90% range holds for 84%; risk labels do not transfer.

The application: GDP estimates where statistics are missing



Click a town: estimate, 90% range and risk label; reveal the official value. Here: the iron-mine town, flagged high risk.

WHAT A USER CAN DO

1. **Estimate any town** from satellite data, with an honest range and a warning when not to trust it.
2. **Compare a new state**: São Paulo predicted vs official, by value or rank, against night lights.
3. **See where satellites are blind**: risk groups and the largest misses, with reasons.



Works offline: `app/index.html`. Live demo now.

What did not work, and what it taught us

WE TRIED	WHAT HAPPENED	LESSON
Night-light steering	No gain once the aggregator was calibrated; an early “steering helps” result vanished	Calibration cut error −21%: fix the basics before adding ingredients
Text alignment	Small gain, shrank in the rerun (−0.025 → −0.009)	Frozen embeddings limit what text can add
POIs and roads	Nothing on top of text	The fact sentences already carry them
Stage C vectors as an “economic space”	Similar-town search matches size, not wealth (R^2 0.22–0.41); clusters explain 0.05–0.14	The vectors work only inside the model; use estimates, not distances
Risk labels in a new state	581 of 645 São Paulo towns flagged high	Ranges travel; warnings need local recalibration

All checks out-of-sample; details in the report and reports/tables (rerun_check, stage_c_twins, clusters).

Answers to the five questions, and what we could not do

RESEARCH QUESTIONS

RQ1 night-light steering	NO CLEAR EFFECT
RQ2 text (captions)	SMALL GAIN, NOT ROBUST
RQ3 POIs and roads	NOTHING ADDED
RQ4 hybrid	BEST, NOT SIGNIFICANT
RQ5 where it fails	MINES, STEEL, COMMUTER TOWNS
Beyond: a new state	SATELLITES TRAVEL, LIGHTS DO NOT

Every answer out-of-sample on regions never seen; 95% block-bootstrap intervals; confirmed by a full rerun.

LIMITS

- GSED is frozen: text can reorganise, not add, image information.
- Caption audit labelled by an LLM, one labeller, no agreement score yet.
- Stage C was revised after the first test results (choices made on validation).
- Trained on one state, transferred to one more; other countries and years untested.
- The tool cannot see mines or industry; it warns instead.

TAKE-AWAY

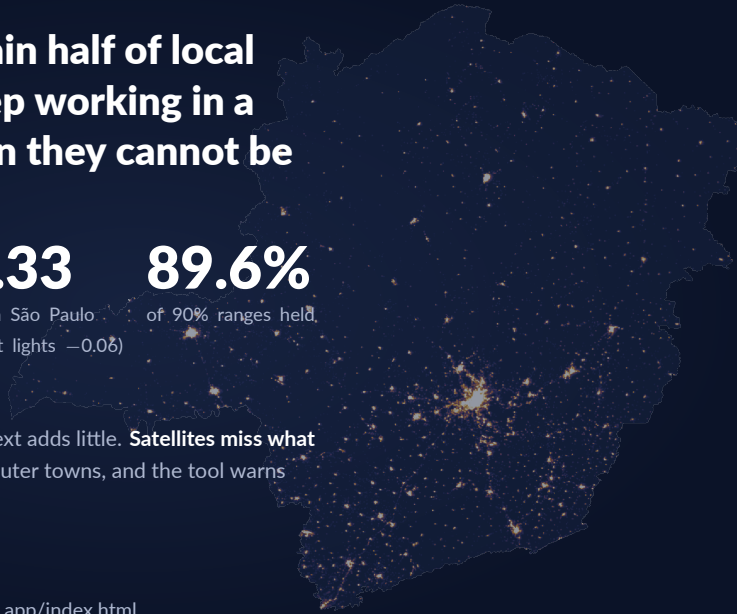
Free satellite data explain half of local income differences, keep working in a new state, and say when they cannot be trusted.

0.52
R^2 in Minas Gerais, unseen regions

0.27
R^2, night lights alone

0.33
R^2 in São Paulo (night lights −0.06)

89.6%
of 90% ranges held

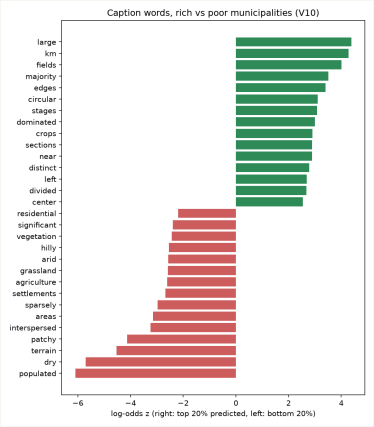


Steering, POIs and roads add nothing clear; text adds little. **Satellites miss what they cannot see:** mines, steel mills and commuter towns, and the tool warns about them.

Questions?

github.com/chadsaras/GIS_Project · live demo: app/index.html

Backup: the text space speaks the state's economic geography



RICHEST 20% PREDICTED

large, *fields*, *circular*, *crops*, *divided*: mechanised farming and centre pivots of the Triângulo

POOREST 20% PREDICTED

sparsely populated, *dry*, *terrain*, *patchy*, *arid*: the semi-arid north and Jequitinhonha

NEAREST TRAINING CAPTIONS

Belo Horizonte (10.82 vs 10.91): dense grids of red-tiled roofs.
Conceição do Mato Dentro (11.94 vs 10.46): forest and pasture, so the mine stays invisible.

Log-odds with informative Dirichlet prior (Monroe et al., 2008); official vs predicted log GDP per capita.

Backup: a full rerun from scratch gives the same answers

Model	R^2 frozen	R^2 rerun	ΔRMSE
V1 night lights	0.271	0.271	0.000
V3 embeddings + C	0.459	0.469	−0.004
V4 + steering	0.441	0.489	−0.020
V6 + text + steering	0.501	0.509	−0.004
V7 full model	0.502	0.519	−0.007
V10 hybrid	**0.523**	**0.522**	0.000
V12 LightGBM	0.386	0.386	0.000

Question	frozen	rerun
RQ1 steering	+0.007	−0.008
RQ2 text	−0.025	−0.009
RQ3 POI, roads	−0.001	−0.004
RQ4 hybrid	−0.009	−0.002

ΔRMSE of each comparison; every 95% interval includes 0 in both runs, so every verdict is unchanged. Deterministic steps match exactly; neural models differ by up to 0.02 RMSE (multi-threaded CPU training).

Backup: Stage C vectors carry economy size, not wealth

TWINS: 10 MOST SIMILAR TRAINING TOWNS

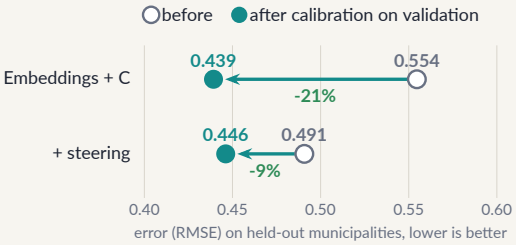
Space	R^2 income/person	R^2 population
Raw satellite vectors	0.31	0.06
Population only	0.00	0.99
Stage C, as trained	0.22–0.33	0.74–0.81
Stage C, size removed	0.28–0.41	0.19–0.47
The model itself	*0.46–0.52*	

CLUSTERS: K-MEANS, HELD-OUT TOWNS

Grouping by	R^2 income/person
Stage C vectors	0.05–0.14
Land cover + lights	0.17–0.23
Raw satellite vectors	0.23–0.29

Stage C predicts total GDP and subtracts population at the end, so its vectors encode size. They are also unstable across trained models (adjusted Rand 0.45 across rounds vs 0.90 for raw vectors).

Backup: calibration mattered more than night-light steering



1. **Predictions were over-spread.** A straight-line correction fitted on validation municipalities cut error by **−21%**.
2. **Honest testing changes conclusions.** Steering looked helpful before the fix; after it, the gain vanished.
3. **Disclosed:** the fix was chosen on validation data, but after the first test results were seen.